Question 1:

Create two new users on your Linux system named 'musicfan' and 'moviebuff', then create a new group called 'playlist' and add both users to this group.

Answer:

- **Step 1: Create the group**

sudo groupadd playlist

- **Step 2: Create the users**

sudo useradd -m -s /bin/bash musicfan

sudo useradd -m -s /bin/bash moviebuff

- **Step 3: Set passwords for the users (optional but recommended)**

sudo passwd musicfan

sudo passwd moviebuff

- **Step 4: Add both users to the 'playlist' group**

sudo usermod -aG playlist musicfan

sudo usermod -aG playlist moviebuff

- **Step 5: Verify group membership**

getent group playlist

(Output should display: playlist:x:<GID>:musicfan,moviebuff)

Question 2:

Inside your home directory, create a folder called 'spotify_shared'. Set its permissions so only users in the 'playlist' group can read and write files inside it, but others cannot access it.

Hint: Use chown and chmod with the group ownership and permission bits.

Answer:

- **Step 1: Create the directory**

mkdir ~/spotify_shared

- **Step 2: Change group ownership to 'playlist'**

sudo chown :playlist ~/spotify_shared

- **Step 3: Set directory permissions**

chmod 770 ~/spotify_shared

(Alternatively, chmod 2770 ~/spotify_shared can be used to set the SGID bit so newly created files automatically inherit the playlist group).

- **Step 4: Verify permissions**

ls -ld ~/spotify_shared

(The output will show drwxrwx--- or drwxrws---, meaning user/owner and playlist group members have read, write, and traverse access, while others have no permissions).

Question 3:

Place a text file named 'top_songs.txt' inside 'spotify_shared'. Change its ownership so that 'musicfan' is the owner and 'playlist' is the group, then set permissions so only the owner can edit, group members can read, and others have no access.

Answer:

- **Step 1: Create the file inside the directory**

touch ~/spotify_shared/top_songs.txt

- **Step 2: Add initial content (optional)**

echo "1. Bohemian Rhapsody - Queen" > ~/spotify_shared/top_songs.txt

- **Step 3: Change ownership to user 'musicfan' and group 'playlist'**

sudo chown musicfan:playlist ~/spotify_shared/top_songs.txt

- **Step 4: Set permissions (Owner: read/write [6], Group: read-only [4], Others: none [0])**

sudo chmod 640 ~/spotify_shared/top_songs.txt

- **Step 5: Verify file permissions**

ls -l ~/spotify_shared/top_songs.txt

(The output will display -rw-r----- 1 musicfan playlist ... top_songs.txt).

Question 4:

Simulate a scenario where 'moviebuff' tries to access 'top_songs.txt'. Switch to the 'moviebuff' user and attempt to open and edit the file. Describe what happens and explain why based on the permissions you set.

Answer:

- **Simulation Commands:**

    1. Switch to user moviebuff:

su - moviebuff

    2. Attempt to read the file:

cat /home/<your-username>/spotify_shared/top_songs.txt

    3. Attempt to edit/append to the file:

echo "2. Hotel California - Eagles" >> /home/<your-username>/spotify_shared/top_songs.txt

(Or open via nano /home/<your-username>/spotify_shared/top_songs.txt and try saving)

- **Observation and Explanation:**
    - **Reading the file:** moviebuff can view and read the file content successfully without errors.
        - *Reason:* moviebuff is a member of the group playlist, and the group permission bit is set to r (read-only).
    - **Editing the file:** The command fails with an error: bash: /home/…/top_songs.txt: Permission denied (or [File is unwritable] in text editors).
        - *Reason:* Write/edit permissions (w) are granted solely to the file owner (musicfan). Since moviebuff is only a member of the group and group write permission is unset, write operations are rejected by the kernel.